

**YeshwantraoChavan College of Engineering***(An Autonomous Institution affiliated to RashtrasantukadojiMaharaj Nagpur University)***(Accredited 'A' Grade by NAAC)**

Hingna Road, Wanadongri, Nagpur - 441 110

Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.edu**E-mail : principal@ycce.edu, info@ycce.edu****Department of Computer Science and Engineering (IoT)****STUDENT'S DIARY/ LOG****Your Name:** Payal SamarKumar Sahu**Your Institute:** Yeshwantrao Chavan College of Engineering Nagpur**Company where internships are carried out :** Cognizant Technology Solutions Pvt. Ltd**Period:** From: 31/01/25 To: 13/05/26

Week: 1		DATE: From	02/02/26 to 06/02/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Behavioural Bootcamp (Icebreaker Week) – Corporate Induction & Soft Skills	
• Attended Corporate Induction sessions covering Cognizant's core values, mission, and work culture. • Participated in Talent Manager Connect and understood the GenC program structure and evaluation framework. • Attended Leader Talks and Alumni interaction sessions to understand Cognizant's BU career pathways. • Got an overview of the QEA (Quality Engineering & Assurance) service line and its role in enterprise software quality. • Understood the 2-stage learning journey: Stage 1 – Core Java with SQL, Stage 2 – Selenium Automation. • Introduced to the Flipped Classroom model and the GenC Learn Platform (Tekstac) for assessments and learning. • Understood the RAG (Red-Amber-Green) Performance Health Status system and interim/final evaluation process. • Attended Behavioural Training (ILT) covering communication, teamwork, and corporate etiquette. • Installed required software: Eclipse IDE 2023-12 and Azul JDK 17 from the Cognizant Company Portal. • Navigated the Tekstac platform using SSO login; explored learning paths and upcoming assessment schedules.			

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Version-2

2025-26



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Fortnightly Progress Report

(To be mailed every alternate Monday to Project Guide, designated person(s) of the company, your HoD and the concerned Dean from your institute)

1. Company Overview:

Cognizant Technology Solutions is a leading global IT services and consulting company headquartered in the USA, with a major delivery presence across India. It specializes in digital transformation, cloud computing, AI, automation, software engineering, and quality assurance. Cognizant's QEA (Quality Engineering and Assurance) division focuses on intelligent and automated quality assurance, modernisation assurance, and experience assurance for enterprise clients across healthcare, finance, and technology sectors.

2. Who did you interact with in this period? Which departments did you visit/work in?

During this first week, I interacted with Cognizant Academy trainers, Talent Managers, and fellow GenC trainees from the QEA Java Track. I attended corporate induction sessions organized by the Academy team and participated in BU Mentor connects and Leader Talks. The interactions were primarily within the training and onboarding division of Cognizant.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Corporate Induction and Program Orientation: Attended sessions explaining Cognizant's core values, the GenC program structure, and the QEA service line. Understood the learning journey, evaluation system (RAG/PHS), and tools like Tekstac and the GenC Learn Platform.

Activity 2 – Behavioural Training: Participated in Instructor-Led Training (ILT) sessions focused on professional communication, teamwork, corporate etiquette, and workplace adaptability. These sessions were designed to build soft skills essential for a corporate IT environment.

Activity 3 – Development Environment Setup: Installed and configured the required software tools including Eclipse IDE 2023-12 and Azul JDK Version 17 from the Cognizant Company Portal, in preparation for the upcoming Core Java training sessions beginning from Week 2.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?



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Faced minor difficulties during software installation on the company-provided system due to access restrictions. With the help of the trainer, the required permissions were obtained and the tools were installed successfully. This experience taught me the importance of seeking timely guidance rather than spending excessive time troubleshooting independently.

5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with the GenC Learn Platform (Tekstac) for accessing learning paths, Udemy course integrations, hands-on exercises, and assessment modules. Also understood the Flipped Classroom model used in the program, where learners study concepts independently before attending trainer-led practice sessions.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

The icebreaker sessions were informative, but allocating a dedicated Q&A slot with past GenC graduates could help new trainees better understand real-world expectations and tips for the upcoming technical training. This would ease the transition from induction to the core technical curriculum.

7. What did you find novel/exciting/valuable during this period?

The most exciting aspect of this week was understanding how Cognizant's QEA division uses AI and automation to transform quality assurance across industries like healthcare and finance. The Leader Talks provided real-world context to the technical skills we will be building over the coming weeks, making the learning journey feel purposeful and industry-aligned.

8. What according to you, can be of use to your institute and students and in what ways?

The structured Flipped Classroom approach combined with the Tekstac platform's auto-evaluated hands-on exercises is a model that could benefit engineering colleges. Introducing students to platforms like Tekstac and agile learning methodologies during their academic years would significantly reduce the adaptation period when they join corporate training programs.



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9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

This being the first week, all planned activities including corporate induction, behavioural training, and environment setup were completed successfully within the allocated time.

Psahu:

Payal SamarKumar Sahu

Name and Signature of Student.

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Week: 2		DATE: From	09/02/26 to 13/02/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Core Java – Stage 1 (Days 1–5): OOP Basics & Java Fundamentals	
• Learned Java basics: OOP principles, JVM/JDK/JRE architecture, and platform independence. • Studied Data Types (primitive & non-primitive), Variable declaration, and Type casting. • Understood OOP concepts: Encapsulation, Inheritance, Polymorphism, and Abstraction. • Practiced Classes & Objects: class structure, object creation, and instance variables. • Studied Constructors: default, parameterized, and constructor overloading. • Completed mandatory Tekstac hands-on: Streamline Your Health Journey, Water Tank Capacity, Square Series. • Practiced Java exercises in Eclipse: basic syntax, OOP programs, and recursion problems. • Attended Master Sessions led by trainer for doubt clarification on Day 1–5 topics. • Completed Udemy Section 2 (Java Programming) from 'Learn Selenium with Java, Cucumber & Frameworks'. • Participated in Behavioural Training (ILT) sessions on Days 3 and 5 alongside technical learning.			

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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 2, I interacted with Cognizant Academy trainers, fellow GenC trainees from the QEA Java Track, and BU Mentors during the Business Mentor Connect session. All sessions were conducted within the Training division. The trainer-led Master Sessions provided guidance on Core Java topics covered in Days 1 to 5.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Core Java ILT Sessions: Attended trainer-led Master Sessions on Java fundamentals including OOP concepts, data types, constructors, and class structures (Chapters 1 and 2 of the BU curriculum).

Activity 2 – Tekstac Hands-On: Completed mandatory hands-on exercises on the Tekstac platform: Streamline Your Health Journey, Water Tank Capacity, and Square Series, which were auto-evaluated.

Activity 3 – Eclipse Practice: Practiced topic-wise Java programs in Eclipse covering basic syntax, OOP, recursion, and factorial programs as per the daily learning guide.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Initially found it difficult to understand the difference between method overloading and method overriding. Referred to the Udemy course Section 2 and clarified doubts during the trainer's Master Session. This reinforced the importance of completing self-paced learning before doubt-clarification sessions.

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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Practiced coding on the Tekstac platform's built-in code editor, where hands-on solutions are automatically evaluated upon compilation. Also explored the Udemy course 'Learn Selenium with Java, Cucumber & Frameworks' (Section 2) as recommended in the daily learning guide.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

More time could be allocated for Eclipse-based practice sessions in the daily schedule. Hands-on coding in a local IDE alongside Tekstac exercises would strengthen problem-solving skills more effectively.

7. What did you find novel/exciting/valuable during this period?

Understanding how OOP principles such as Encapsulation, Inheritance, and Polymorphism form the backbone of real-world software development was the most valuable insight of this week. The Flipped Classroom approach made learning more engaging.

8. What according to you, can be of use to your institute and students and in what ways?

The practice of completing topic-wise hands-on exercises on an auto-evaluated platform like Tekstac immediately after learning each concept is highly effective. Engineering colleges could adopt similar LMS-based auto-graded exercises to improve programming skill development.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 2 were completed. The mandatory Tekstac hands-on exercises, Eclipse practice programs, and Udemy Section 2 learnings were all finished within the scheduled time.

Payal SamarKumar Sahu
Name and Signature of Student.

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Week: 3		DATE: From	16/02/26 to 20/02/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Core Java – Stage 1 (Days 6–10): Collections, Exception Handling & Java 8 Features	
• Studied Exception Handling: try-catch block, multiple catch, finally, throw and throws keywords. • Learned Java Collections Framework: ArrayList, LinkedList, HashSet, HashMap, TreeSet, TreeMap. • Understood Set Interface (HashSet, LinkedHashSet, TreeSet) and Map Interface (HashMap, LinkedHashMap). • Studied Iterator and ListIterator for traversing collections. • Learned Java 8 Features: Lambda Expressions, Stream API (filter, map, reduce), and Optional class. • Understood Generics: generic classes, methods, and wildcards; Annotations (@Override, @Deprecated). • Studied Design Patterns: Singleton, Factory, Observer, and Strategy patterns. • Completed mandatory Tekstac hands-on: WordZee, Fishing Competition, Brilliant Restaurant, Investment Calculation. • Practiced Eclipse exercises on Arrays, String handling, Collections, and Exception handling. • Attended Business Mentor Connect session and Practice ICT Assessment on Day 7 (17/02/26).			

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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 3, I interacted with Cognizant Academy trainers, fellow GenC trainees, and a BU Mentor during the Business Mentor Connect session held on Day 7 (17/02/26). All interactions took place within the Training division. The trainer-led Master Sessions covered Core Java advanced topics for Days 6 to 10.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Java Collections & Exception Handling (ILT): Attended Master Sessions covering the Java Collections Framework (ArrayList, HashMap, HashSet, TreeMap) and Exception Handling (try-catch, finally, custom exceptions). Practiced all examples in Eclipse.

Activity 2 – Tekstac Hands-On Exercises: Completed mandatory hands-on problems – WordZee, Fishing Competition, Brilliant Restaurant, and Investment Calculation – on the Tekstac platform with auto-evaluation.

Activity 3 – Business Mentor Connect & ICT Practice: Participated in the BU Mentor Connect session on Day 7 for industry insights. Also attempted the Practice ICT Assessment on the same day to gauge understanding of Java concepts covered so far.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Found it challenging to understand the differences between HashMap, LinkedHashMap, and TreeMap in terms of ordering and performance. Clarified doubts during the Master Session and referred to the Udemy course and GeeksForGeeks resources. This helped in understanding when to use each collection type based on the use case.



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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Practiced Java Collections and Exception Handling on the Tekstac platform. Additionally, explored GeeksForGeeks resources on Serial and Parallel Sorts in Java and Java 8 Stream API as recommended in the learning guide. Also referred to the Udemy course 'Java In-Depth: Become a Complete Java Engineer' (Sections 16 and 17) for Collections Framework and Generics.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

Including more real-world use-case examples while explaining Design Patterns (Singleton, Factory) during Master Sessions would help trainees better understand when and why to use each pattern in enterprise software development.

7. What did you find novel/exciting/valuable during this period?

Learning Java 8 features such as Lambda Expressions and the Stream API was the most exciting part of this week. Understanding how streams allow functional-style operations on collections made code much cleaner and more readable. The Business Mentor Connect session also provided valuable industry perspective on how these concepts are applied in real projects.

8. What according to you, can be of use to your institute and students and in what ways?

Teaching Java Collections Framework and Java 8 features (Streams, Lambda) as part of the academic Java curriculum would greatly benefit students. These are industry-standard topics that are heavily used in enterprise applications, and early exposure would reduce the learning curve when students join corporate training programs.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 3 were completed successfully. The mandatory Tekstac hands-on exercises, Eclipse practice programs, Udemy course sections, and the Practice ICT Assessment were all finished within the scheduled time.

Psahu.

Payal SamarKumar Sahu

Name and Signature of Student.

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Week: 4		DATE: From	23/02/26 to 27/02/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Core Java – Stage 1 (Days 11–15): Java 8, JDBC, Code Challenge & ICT Assessment	
• Studied Java 8 Features: Streams, Optionals, Asynchronous and Parallel Programming concepts. • Learned JDBC (Java Database Connectivity): architecture, DriverManager, Statement, PreparedStatement. • Practiced JDBC CRUD operations: Insert, Update, Delete, Select queries and Transactions (Commit, Rollback). • Completed mandatory Tekstac hands-on: Mystic Line, Area and Volume, The Perfect Password, Library Management. • Attended Udemy course – Java Database Connection: JDBC and MySQL (complete course). • Completed Java Code Challenge Assessment (Group 1 & Group 2) – 2 hrs, 3 attempts, 1 question each. • Attempted Java ICT Assessment (Core Java Skill War – Skill Based Assessment [101-Basics]) on Day 15. • Practiced topic-wise Eclipse hands-on: String Handling, Collections Framework, and File Handling programs. • Revised all Core Java topics (Days 1–15) in preparation for Stage 1 qualifier assessment. • Attended Behavioural Training (ILT) sessions on Days 11 and 13 alongside technical learning.			

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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 4, I interacted with Cognizant Academy trainers and fellow GenC trainees from the QEA Java Track. The trainer-led Master Sessions covered Java 8 features and JDBC for Days 11 to 15. Interactions were primarily within the Training division, with additional engagement during the Code Challenge and ICT Assessment sessions.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – JDBC Learning & Practice: Attended ILT sessions on JDBC architecture, CRUD operations, and transactions. Completed the full Udemy course 'Java Database Connection: JDBC and MySQL' and practiced examples in Eclipse.

Activity 2 – Code Challenge Assessment: Attempted the Java Code Challenge Assessment (Group 1 & Group 2) on Days 11 and 12. Each attempt had 1 question, 2-hour duration, and 3 attempts allowed. This helped benchmark coding proficiency.

Activity 3 – Java ICT Assessment: On Day 15, appeared for the Core Java Skill War – Skill Based Assessment (4-hour duration, 3 attempts, 1 question per attempt). This was the final assessment for Stage 1 Java skills.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Faced difficulty in writing optimized SQL queries within JDBC for multi-table operations. Revisited the Udemy JDBC course sections on PreparedStatement and joins, and practiced additional examples in Eclipse. This improved both my SQL query writing and JDBC integration skills significantly.



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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with JDBC architecture and the process of connecting a Java application to a MySQL database using DriverManager, Connection, Statement, and PreparedStatement interfaces. Also explored Java 8 Stream API and Asynchronous Programming concepts through GeeksForGeeks reference links provided in the learning guide.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

Providing a sample mini-project integrating JDBC with Java OOP concepts as a reference during the JDBC sessions would help trainees better understand how database connectivity is applied in real enterprise applications, rather than learning it in isolation.

7. What did you find novel/exciting/valuable during this period?

The Java ICT Assessment (Core Java Skill War) was a challenging yet valuable experience. Solving a complete problem using all the Java concepts learned over 15 days under a timed environment greatly improved confidence and problem-solving speed. The Code Challenge assessments also helped identify individual strengths and areas needing improvement.

8. What according to you, can be of use to your institute and students and in what ways?

Introducing JDBC and database connectivity as part of the academic Java lab curriculum would be highly beneficial. Many students are unaware of how Java connects to databases in enterprise systems. Including this with hands-on projects would give students a significant advantage when they start corporate training or internships.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 4 were completed. The Tekstac hands-on exercises, JDBC Udemy course, Code Challenge Assessments, and the Java ICT Assessment were all attempted and completed within the scheduled time as per the learning guide.

Psahu:

Payal SamarKumar Sahu

Name and Signature of Student.

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Week: 5		DATE: From	02/03/26 to 06/03/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		SQL – Stage 1 (Days 16–20): SQL Fundamentals, Code Challenges & Stage 1 Assessment	
• Learned SQL fundamentals: DDL (CREATE, ALTER, DROP), DML (INSERT, UPDATE, DELETE), and SELECT queries. • Studied Relational Database Design concepts and NoSQL overview through UdeMy and reference links. • Practiced SQL JOINS (INNER, LEFT, RIGHT), Subqueries, Aggregate Functions, and String/Date Functions. • Completed mandatory Tekstac hands-on: Create Borrowing Table, Alter Table, Insert/Update/Delete Records. • Completed SQL Code Challenge Assessments: DDL/DML & Select, JOINS & Subqueries, Scalar & Aggregate. • Learned XML and JSON basics through UdeMy course sections on API Technical Writing. • Attempted Stage 1 Mock Assessment (3 hrs, 3 attempts, 3 questions) on Day 17 (03/03/26). • Appeared for Stage 1 Final Assessment (3 hrs, 3 SBA Java questions) on Days 20–21 (06/03/26). • Attended Business Mentor Connect session on Day 17 alongside the Stage 1 Mock Assessment. • • Attended Behavioural Training (ILT) sessions on Days 17 and 18 alongside SQL learning.			

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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 5, I interacted with Cognizant Academy trainers, fellow GenC trainees from the QEA Java Track, and a BU Mentor during the Business Mentor Connect session on Day 17. All activities were conducted within the Training division. The trainer-led Master Sessions covered SQL concepts from Days 16 to 19, and the Stage 1 final assessment was held on Day 20.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – SQL Learning (ILT & Self-Paced): Attended trainer-led Master Sessions on SQL covering DDL, DML, SELECT queries, JOINS, Subqueries, Aggregate Functions, and String/Date Functions. Also completed the Udemy course 'SQL for Beginners: Learn SQL using MySQL and Database Design' (Sections 1–14).

Activity 2 – SQL Code Challenge Assessments: Completed three SQL Code Challenge Assessments on Days 18 and 19 covering DDL/DML & Select statements, JOINS and Subqueries, and Scalar & Aggregate functions. Each was 2 hours long with 3 attempts.

Activity 3 – Stage 1 Assessment: Appeared for the Stage 1 Final Assessment on Day 20 (06/03/26). The assessment was 3 hours long with 3 SBA Java questions covering all Stage 1 skills. Score from the first instance was considered final.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Found subqueries with multiple levels of nesting challenging during the Code Challenge. Referred to the Udemy course Section 11 (Subqueries) and practiced additional examples from the Tekstac platform. This helped in understanding how to break complex queries into smaller logical parts before combining them.



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Department of Computer Science and Engineering (IoT)

5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with MySQL database management through the Udemy course on SQL for Beginners and the Relational Database Design course. Also explored the RANK function and NoSQL overview through reference web pages provided in the learning guide. Practiced all SQL queries in the Tekstac platform's built-in editor.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

Providing a structured set of practice SQL problems covering all difficulty levels – basic, intermediate, and advanced – before the Code Challenge Assessments would help trainees better prepare. Currently, the jump from hands-on exercises to Code Challenge questions feels slightly large.

7. What did you find novel/exciting/valuable during this period?

Appearing for the Stage 1 Final Assessment was the most significant milestone of this week. It tested the combined knowledge of Core Java and SQL acquired over the past four weeks. The experience of solving real-world-style SBA questions under time pressure was both challenging and exciting, and built confidence for the upcoming Stage 2 modules.

8. What according to you, can be of use to your institute and students and in what ways?

Teaching SQL along with Java in academic courses and introducing students to database connectivity (JDBC) early on would be highly beneficial. Most engineering students learn SQL and Java separately without understanding their integration. A combined lab project using Java with MySQL would bridge this gap effectively.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 5 were completed. The Tekstac SQL hands-on exercises, SQL Code Challenge Assessments, Stage 1 Mock Assessment, and the Stage 1 Final Assessment were all attempted and completed within the scheduled time as per the learning guide.

Psahu

Payal SamarKumar Sahu

Name and Signature of Student.

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering (IoT)****STUDENT'S DIARY/ LOG****Your Name:** Payal SamarKumar Sahu**Your Institute:** Yeshwantrao Chavan College of Engineering Nagpur**Company where internships are carried out :** Cognizant Technology Solutions Pvt. Ltd**Period:** From: 31/01/25 To: 13/05/26

Week: 6		DATE: From	09/03/26 to 13/03/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Stage 2 – Functional Testing (Days 22–25): SDLC, STLC, Agile, Scrum & JIRA Tool	
• Understood SDLC and STLC phases: Use case, Test scenario, Test case, Test Environment, Execution, Defect Reporting, Test Closure, and Test Metrics. • Learned testing types: White Box vs Black Box, Static vs Dynamic, Verification vs Validation, Functional vs Non-Functional. • Studied test design techniques including Regression Testing, Smoke Testing, Sanity Testing, Exploratory Testing, and End-to-End Testing. • Understood Agile and Scrum methodologies: Scrum Teams, Sprints, User Stories, Definition of Done, Sprint Planning, Review and Retrospective. • Learned JIRA Tool: installation, project creation, user management, Backlog, Epic, User Stories, Sprint creation, and Bug reporting. • Practiced writing Test Cases and Test Scenarios; understood Defect Management and Test Sign-off process. • Completed mandatory Tekstac hands-on: The Importance of QEA and Career Scope, Scope Debate activity. • Started Trainer-Guided Integrated Project: Activity Readiness, Configured JIRA Project and Backlog Management. • Attended Business Mentor Connect session on Day 22 (10/03/26) for industry exposure.			

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Fortnightly Progress Report

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1. Company Overview:

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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 6, I interacted with Cognizant Academy trainers, fellow GenC trainees from the QEA track, and a BU Mentor during the Business Mentor Connect session on Day 22 (10/03/26). All sessions were conducted within the Training division. The trainer-led Master Sessions covered Functional Testing, Agile/Scrum methodologies, and JIRA tool usage for Days 22 to 25.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Functional Testing & STLC (ILT): Attended Master Sessions covering SDLC, STLC phases, testing types (White Box, Black Box, Regression, Smoke, Sanity), and test design techniques. Completed the Udemmy course 'Learn Manual Software Testing + Agile with Jira Tool' covering all modules.

Activity 2 – JIRA Tool Hands-On: Installed and configured JIRA, created a project, managed Backlog, Epics, and User Stories, created Sprints, and practiced Bug creation and Test Case writing using the Zephyr plugin as part of the Trainer-Guided Integrated Project.

Activity 3 – Integrated Project Initiation: Began the Trainer-Guided Integrated Project by completing Activity Readiness and setting up JIRA Project and Backlog Management. This project will run in parallel with all subsequent learning modules through Stage 2.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Initially found JIRA's Sprint lifecycle and Zephyr plugin configuration complex. Referred to the Udemmy course sections on JIRA Tool Implementation and sought clarification during the Master Session. This helped in understanding the end-to-end test management workflow within JIRA effectively.



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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with the JIRA project management tool for Agile test management including Sprint creation, Backlog grooming, Bug tracking, and Test Case execution using the Zephyr plugin. Also studied the complete STLC process from requirement analysis to test closure, and explored how Agile methodology integrates testing within each sprint.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

A live demo session showing how a real-world project moves through all STLC phases within JIRA – from requirements to defect closure – would greatly help trainees visualize the complete testing lifecycle. Currently, each phase is taught independently, and connecting them through a single live project scenario would improve understanding.

7. What did you find novel/exciting/valuable during this period?

Understanding the Agile and Scrum framework in depth was the most valuable experience this week. Learning how Sprint Planning, Backlog Refinement, Sprint Review, and Retrospective ceremonies work together to deliver software iteratively gave great insight into how modern IT projects are managed. The JIRA hands-on further reinforced these concepts practically.

8. What according to you, can be of use to your institute and students and in what ways?

Introducing JIRA and Agile methodology as part of the academic project management or software engineering curriculum would be extremely beneficial. Most engineering students are unaware of industry-standard project tracking tools. Even a basic hands-on session with JIRA during academic project work would prepare students much better for corporate environments.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 6 were completed. The Functional Testing Udemy course, JIRA hands-on exercises, Tekstac mandatory hands-on, Trainer-Guided Integrated Project initial deliverables, and the Business Mentor Connect session were all completed within the scheduled time.

Payal SamarKumar Sahu

Name and Signature of Student.

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering (IoT)****STUDENT'S DIARY/ LOG****Your Name:** Payal SamarKumar Sahu**Your Institute:** Yeshwantrao Chavan College of Engineering Nagpur**Company where internships are carried out :** Cognizant Technology Solutions Pvt. Ltd**Period:** From: 31/01/25 To: 13/05/26

Week: 7		DATE: From	16/03/26 to 20/03/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Stage 2 – Functional Testing ICT Assessment & Selenium Automation Basics (Days 26–27)	
• Appeared for Functional Testing ICT Assessment (Day 26, 16/03/26) – 4 hrs, 3 attempts, 1 question per attempt. • Completed Functional Testing Skill Based Assessment covering STLC, Agile, JIRA, and test design techniques. • Began Stage 2 Selenium Automation: studied Selenium basics – history, components (WebDriver, IDE, Grid), and supported languages. • Understood Selenium WebDriver architecture: WebDriver API, RemoteWebDriver, ChromeDriver, FirefoxDriver, EdgeDriver. • Learned Selenium environment setup: installing JDK, Eclipse IDE, and configuring Selenium WebDriver with Eclipse. • Studied WebDriver locators: finding elements by Id, Name, LinkText, PartialLinkText, ClassName, CSS, and TagName. • Completed mandatory Tekstac hands-on: Find Element by Id, Find Element by Name. • Completed Udemy course Section 3 (Selenium WebDriver) – Subtopics 17 to 25. • Started Integrated Project Deliverable: Automation Feasibility Report and Strategy (due by Day 31). • Started GenAI learning: Fundamentals of Generative AI, Introduction to Responsible AI, and GitHub Copilot basics.			

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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 7, I interacted with Cognizant Academy trainers and fellow GenC trainees from the QEA Selenium Automation track. Day 26 was focused on the Functional Testing ICT Assessment, while Days 27 onwards involved trainer-led Master Sessions on Selenium WebDriver basics. All interactions were within the Training division

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Functional Testing ICT Assessment: Appeared for the Functional Testing Integrated Capability Test on Day 26 (16/03/26). The assessment was 4 hours long with 3 attempts and 1 question per attempt, covering all Functional Testing skills including STLC, Agile, and JIRA.

Activity 2 – Selenium WebDriver Setup & Locators: Attended Master Sessions on Selenium basics, WebDriver architecture, and environment configuration. Practiced finding web elements using locators such as Id, Name, ClassName, CSS Selector, and XPath in Eclipse with a live browser.

Activity 3 – GenAI Learning & Integrated Project: Started the GenAI module covering Fundamentals of Generative AI and Responsible AI. Also initiated the Integrated Project Deliverable – Automation Feasibility Report and Strategy as part of the parallel project work.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Faced difficulty setting up Selenium WebDriver with the correct ChromeDriver version compatible with the installed Chrome browser. Resolved this by downloading the matching ChromeDriver version from the official site and configuring the system path correctly as guided by the trainer. This taught the importance of version compatibility in automation tool setup.



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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with the Selenium WebDriver architecture and the process of locating web elements using different locator strategies. Also explored the GenC Learn Platform's BU Training Content for Chapter 2 (Selenium Architecture) and Chapter 3 (Setting up Selenium Environment) including PPT and video learning reference content as provided in the learning guide.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

A pre-configured virtual machine or Docker image with all Selenium dependencies already set up would save significant setup time during the initial days of Selenium training. Environment configuration issues consumed considerable time that could have been used for actual hands-on practice.

7. What did you find novel/exciting/valuable during this period?

Writing the very first Selenium WebDriver script that launched a browser, navigated to a URL, and located a web element using an Id locator was the most exciting experience of this week. Seeing the automation script control the browser in real time made the concept of test automation come alive and sparked great enthusiasm for the upcoming modules.

8. What according to you, can be of use to your institute and students and in what ways?

Introducing Selenium WebDriver as part of the academic software testing curriculum would be very valuable. Most students have theoretical knowledge of testing but no exposure to automation tools. Even a basic hands-on lab on browser automation using Selenium with Java would give students a significant head start when they join industry training programs.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 7 were completed. The Functional Testing ICT Assessment, Selenium environment setup, Tekstac hands-on exercises (Find Element by Id and Find Element by Name), Udemy Section 3 (Subtopics 17–25), and the initial Integrated Project deliverable were all completed within the scheduled time.

Payal SamarKumar Sahu

Name and Signature of Student.

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering (IoT)****STUDENT'S DIARY/ LOG****Your Name:** Payal SamarKumar Sahu**Your Institute:** Yeshwantrao Chavan College of Engineering Nagpur**Company where internships are carried out :** Cognizant Technology Solutions Pvt. Ltd**Period:** From: 31/01/25 To: 13/05/26

Week: 8		DATE: From	23/03/26 to 27/03/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Stage 2 – Selenium WebDriver Locators, Dynamic XPath & Automation Techniques (Days 28–32)	
• Studied advanced WebDriver locators: CSS Selector (Tag, ID, Class, Attribute), Relative and Chained Locators, and Handling Shadow DOM in Selenium. • Learned Dynamic XPath: Single Slash, Double Slash, contains(), starts_with(), text(), and Last() functions for locating dynamic web elements. • Understood Desired Capabilities in Selenium WebDriver for browser configuration. • Practiced WebDriver commands: text fields, buttons, FindElement vs FindElements, handling Canvas elements, and JavaScript Executor. • Learned WebDriver Actions class: mouse and keyboard actions, drag and drop, handling frames, iFrames, multiple windows, and tool tips. • Studied WebDriver Waits: Implicit Wait, Explicit Wait, and Fluent Wait for handling dynamic elements. • Completed mandatory Tekstac hands-on: Find options in dropdown, Find Radio Element, Find Element by TagName, Handling CheckBox, LinkText, PartialLinkText, CSSLocator, AbsoluteXpathLocator, xpathAncestor, Working with Alerts, JavaScript Executor. • Completed Integrated Project Deliverable: Configured Automation Environment & Project Structure (due Day 31). • Completed Selenium WebDriver Code Challenge Assessment – 2 hrs, 3 attempts, 1 question per attempt. • Continued Mini Project work and GenAI Code Assist (GitHub Copilot) learning alongside daily sessions.			

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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 8, I interacted with Cognizant Academy trainers and fellow GenC trainees from the QEA Selenium Automation track. Trainer-led Master Sessions covered advanced WebDriver locators, Dynamic XPath, and WebDriver Actions for Days 28 to 32. All interactions were within the Training division, with additional engagement during the Code Challenge Assessment on Day 31.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Dynamic XPath & Advanced Locators (ILT): Attended Master Sessions on Dynamic XPath techniques including contains(), starts_with(), text(), and Last() functions. Practiced locating dynamic web elements that change their attributes on every page load, which is a critical real-world automation skill.

Activity 2 – WebDriver Actions & Waits: Learned and practiced the Actions class for mouse/keyboard interactions, drag-and-drop, handling frames, iFrames, alerts, and multiple windows. Also studied the three types of WebDriver Waits – Implicit, Explicit, and Fluent – for synchronizing test scripts with dynamic page behaviour.

Activity 3 – Selenium Code Challenge & Integrated Project: Completed the Selenium WebDriver Code Challenge Assessment on Day 31. Also submitted the Integrated Project Deliverable for Configured Automation Environment and Project Structure as part of the ongoing parallel project work.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Faced difficulty writing reliable Dynamic XPath expressions for elements whose attributes change on page refresh. Practiced multiple XPath strategies using contains() and starts_with() functions on the Tekstac platform and the BU Chapter hands-on exercises. This reinforced the importance of using partial attribute matching for robust element identification in real-world automation.



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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with the BU Training Content chapters 4 to 7 covering Selenium WebDriver, Locating Web Elements, Desired Capabilities, and WebDriver commands. Also explored the Tekstac platform's real-time hands-on scenarios for Relative Locators, Chained Locators, Shadow DOM handling, and JavaScript Executor execution as provided in the Best Practices section of the learning guide.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

Having a dedicated practice web application (AUT – Application Under Test) hosted internally for all Selenium hands-on exercises would be more consistent than using live public websites that may change their structure unexpectedly. This would ensure that all locators and XPath expressions remain valid throughout the training period.

7. What did you find novel/exciting/valuable during this period?

Learning Dynamic XPath and how it can be used to locate elements that have no fixed attributes was the most valuable insight of this week. Understanding how `contains()`, `starts_with()`, and `text()` functions work in XPath expressions opened up a completely new dimension of element identification. The Shadow DOM handling concept was also novel as it is a relatively advanced browser concept rarely covered in academic courses.

8. What according to you, can be of use to your institute and students and in what ways?

Introducing Selenium WebDriver as part of the academic software testing curriculum would be very valuable. Most students have theoretical knowledge of testing but no exposure to automation tools. Even a basic hands-on lab on browser automation using Selenium with Java would give students a significant head start when they join industry training programs.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 8 were completed. The mandatory Tekstac hands-on exercises, BU Chapter content (Chapters 4–7), Udemy Section 3 (Subtopics 26–32), Selenium WebDriver Code Challenge Assessment, and the Integrated Project Deliverable for Configured Automation Environment were all finished within the scheduled time.

Payal SamarKumar Sahu

Name and Signature of Student.

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering (IoT)****STUDENT'S DIARY/ LOG****Your Name:** Payal SamarKumar Sahu**Your Institute:** Yeshwantrao Chavan College of Engineering Nagpur**Company where internships are carried out :** Cognizant Technology Solutions Pvt. Ltd**Period:** From: 31/01/25 To: 13/05/26

Week: 9		DATE: From	30/03/26 to 03/04/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Stage 2 – Selenium WebDriver with POM, Apache POI & Mini Project (Days 33–39)	
• Studied Page Object Model (POM) design pattern: structure, benefits, separation of test logic from page elements. • Learned Page Factory Model: @FindBy annotations, initElements() method, and creating reusable page classes. • Practiced Apache POI: Excel Read Operation, Excel Write Operation, and Applying POI with POM for Flight Reservation. • Understood Data-Driven Testing using Apache POI – reading test data from Excel sheets for parameterized test execution. • Studied Selenium DatePicker using Actions Class and Calendar-based automation scenarios. • Completed mandatory Tekstac hands-on: Excel Read Operation, Excel Write Operation, Applying POI – Flight Reservation, POM with Page Factory Model. • Completed BU Chapter 12 hands-on (Apache POI) and Chapter content for Selenium WebDriver commands (Chapters 7–11). • Completed Selenium WebDriver with POM and ApachePOI Code Challenge Assessment – 2 hrs, 3 attempts. • Submitted Integrated Project Deliverable: Refactored Automation Framework (POM Implementation) and Data Management Strategy (Apache POI) – due Day 41. • Continued Mini Project integration – automated individual functional scenarios and submitted for SME review.			

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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 9, I interacted with Cognizant Academy trainers and fellow GenC trainees from the QEA Selenium track. Trainer-led Master Sessions covered POM design pattern, Apache POI data-driven testing, and advanced WebDriver techniques for Days 33 to 39. All interactions were within the Training division, with additional engagement during the Code Challenge Assessment on Day 39.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Page Object Model (POM) Implementation: Attended Master Sessions on the POM design pattern and Page Factory Model. Practiced creating separate Page classes with `@FindBy` annotations and `initElements()` in Eclipse, separating test logic from UI element definitions for better maintainability.

Activity 2 – Apache POI Data-Driven Testing: Learned to use the Apache POI library to read from and write to Excel files within Selenium test scripts. Practiced Excel Read and Write operations and integrated POI with the POM framework for the Flight Reservation scenario on the Tekstac platform.

Activity 3 – Mini Project Integration & Code Challenge: Submitted the Mini Project deliverables including automated test scripts for individual functional scenarios. Also completed the Selenium WebDriver with POM and ApachePOI Code Challenge Assessment on Day 39 to benchmark skills in these areas.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Initially faced difficulty managing file paths for Excel files in the Apache POI read/write operations across different system environments. Resolved this by using relative paths and the `ClassLoader.getResource()` method as suggested during the Master Session. This reinforced best practices for writing environment-independent automation code.



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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with the Page Object Model (POM) design pattern and its implementation using Page Factory in Selenium. Also explored the Apache POI library for Excel-based data-driven testing, and studied BU Training Content for Chapters 8 to 12 covering WebDriver Actions, Multiple Windows, WebDriver Waits, Selenium Automation Techniques, and Apache POI with their respective hands-on, PPT, and video learning reference content.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

Providing a pre-built sample POM project structure as a reference template at the start of the POM module would help trainees get started faster. Currently, building the project structure from scratch during learning takes extra time. Having a reference project to compare against would accelerate understanding and reduce errors.

7. What did you find novel/exciting/valuable during this period?

Implementing the Page Object Model for the first time and seeing how it transforms messy, hardcoded test scripts into a clean, organized, and reusable framework was the most exciting experience of this week. The concept of separating test logic from page elements using Page classes and Page Factory annotations gave a real sense of how professional automation frameworks are built in industry.

8. What according to you, can be of use to your institute and students and in what ways?

Teaching the Page Object Model as part of academic software testing or Java projects would be highly beneficial. POM is an industry-standard design pattern used in virtually every enterprise Selenium framework. Introducing students to design patterns in the context of test automation would bridge the gap between academic Java knowledge and its real-world application in QA engineering.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 9 were completed. The Tekstac hands-on exercises for POM and Apache POI, BU Chapter content (Chapters 7–12), Udemy Section 4 (Apache POI – Data Driven Testing), the Code Challenge Assessment, Integrated Project deliverables, and Mini Project integration were all completed within the scheduled time.

Payal SamarKumar Sahu

Name and Signature of Student.

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering (IoT)****STUDENT'S DIARY/ LOG****Your Name:** Payal Samarkumar Sahu**Your Institute:** Yeshwantrao Chavan College of Engineering Nagpur**Company where internships are carried out :** Cognizant Technology Solutions Pvt. Ltd**Period:** From: 31/01/25 To: 13/05/26

Week: 10		DATE: From	06/04/26 to 10/04/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Stage 2 – Selenium with TestNG, ICT Assessments & Interim Evaluation (Days 40–45)	
• Learned TestNG framework: Introduction, creating TestNG test cases, annotations (@Test, @BeforeMethod, @AfterMethod, @BeforeSuite, @AfterSuite) and assertions. • Studied TestNG concepts: Data-Driven Testing using @DataProvider, running tests in parallel, setting test priority, and dependency between tests. • Practiced Soft and Hard Assertions in TestNG, multi-browser testing, and TestNG Report Generation (PDF and Email reports). • Learned Data-Driven Testing from XML and JSON files: practiced XMLParsing and JSONParsing hands-on exercises. • Completed Selenium with TestNG Code Challenge Assessment – 2 hrs, 3 attempts, 1 question per attempt (Day 41). • Attempted Selenium Mock ICT Assessment (Day 42) and Selenium ICT Assessment (Day 43) – 4 hrs, 3 attempts each. • Submitted Integrated Project Deliverable: Integrated Test Suite using TestNG Framework, TestNG DataProviders, testng.xml, Extent Reports and Listeners (due Day 43). • Delivered Mini Project to SME for review and incorporated feedback comments. • Appeared for Interim Project Evaluation + Interim Technical Evaluation via video interview on Days 44–45 (09–10/04/26).			

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Fortnightly Progress Report

(To be mailed every alternate Monday to Project Guide, designated person(s) of the company, your HoD and the concerned Dean from your institute)

1. Company Overview:

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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 10, I interacted with Cognizant Academy trainers and fellow GenC trainees during daily Master Sessions on TestNG. On Days 44 and 45, I was interviewed by a Technical Subject Matter Expert (SME) from the Business Unit during the Interim Technical and Project Evaluation. This was conducted via a video interview on the GenC Learn Platform, representing my first formal evaluation interaction with a BU professional.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – TestNG Framework Learning & Hands-On: Attended ILT Master Sessions covering TestNG annotations, assertions, data providers, parallel execution, XML/JSON data-driven testing, and report generation. Completed all mandatory Tekstac hands-on exercises including XMLParsing and JSONParsing scenarios.

Activity 2 – Selenium ICT Assessments: Attempted the Selenium Mock ICT on Day 42 and the actual Selenium ICT Assessment on Day 43. Both were 4-hour assessments with 1 question per attempt designed to test integrated Selenium skills including POM, Apache POI, and TestNG concepts.

Activity 3 – Interim Evaluation (Project + Technical): On Days 44–45, appeared for the Interim Project Evaluation where the Mini Project was reviewed by the BU SME, followed by the Interim Technical Evaluation via video interview covering all skills learned up to Stage 2 halfway point. There was only one attempt for the interim evaluation.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Found the Interim Technical Evaluation interview nerve-wracking as it was the first interaction with a BU SME in a formal assessment setting. Prepared thoroughly by revising all topics from Stage 1 and Stage 2 up to TestNG, and practiced explaining concepts clearly. This experience greatly improved technical communication skills and ability to articulate solutions under pressure.



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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with the TestNG framework for Selenium test management including testng.xml configuration for suite-level execution, grouping tests, and setting up parallel execution across browsers. Also explored Extent Reports integration with TestNG listeners for generating detailed HTML test execution reports, and studied XML and JSON data parsing for data-driven test execution.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

Conducting a mock Interim Evaluation session a few days before the actual evaluation would greatly help trainees understand the format, reduce anxiety, and perform better. Currently the Mock Interview session is mentioned in the learning guide but more structured preparation guidance with sample questions aligned to the BU SME evaluation criteria would be beneficial.

7. What did you find novel/exciting/valuable during this period?

The Interim Project and Technical Evaluation was the most significant milestone experience of the entire training so far. Being evaluated by a real BU SME on both technical knowledge and project work gave a genuine feel of a professional performance review. The feedback received during the evaluation was extremely valuable for identifying areas to strengthen before the Final Evaluation.

8. What according to you, can be of use to your institute and students and in what ways?

The TestNG framework with its powerful annotations, parallel execution, and reporting capabilities is widely used in enterprise automation. Introducing TestNG alongside JUnit in academic Java testing courses would give students hands-on experience with industry-standard test management. The concept of data-driven testing using XML and JSON files could also be incorporated into academic project work to improve test coverage awareness.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 10 were completed. The mandatory Tekstac hands-on exercises, TestNG Code Challenge, Selenium Mock ICT, Selenium ICT Assessment, Integrated Project deliverables for TestNG framework, Mini Project delivery, and the Interim Project and Technical Evaluation were all completed within the scheduled time.

Psahu.

Payal SamarKumar Sahu

Name and Signature of Student.

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering (IoT)****STUDENT'S DIARY/ LOG****Your Name:** Payal SamarKumar Sahu**Your Institute:** Yeshwantrao Chavan College of Engineering Nagpur**Company where internships are carried out :** Cognizant Technology Solutions Pvt. Ltd**Period:** From: 31/01/25 To: 13/05/26

Week: 11		DATE: From	13/04/26 to 17/04/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Stage 2 – Automation Framework & DevOps: GIT, Maven, Selenium Grid, Jenkins (Days 46–51)	
• Studied Automation Framework concepts: Data-Driven, Keyword-Driven, Page Factory, Hybrid Framework design combining POM + Data-Driven + Keyword-Driven approaches. • Learned Test Automation Framework components: Test Data Handling (Excel, CSV, XML, JSON), Test Report and Test Log generation using Extent Reports and Allure Reports. • Understood Custom Logging with Log4j in Selenium and Screenshot Capture Strategies for failure analysis. • Learned GIT and GitHub: basic commands (clone, add, commit, push, pull), branching and merging strategies, and pushing automation framework to remote repository. • Studied Maven: Introduction, creating and managing Maven projects, Maven Build Lifecycle (clean, compile, test, package), managing dependencies and plugins, and executing tests using Maven goals. • Understood Jenkins CI: setting up Jenkins CI server, creating Jenkins Pipeline, integrating GIT and Maven projects, and scheduling/triggering automation runs. • Studied Selenium Grid: architecture (Hub and Node), setting up and configuring Grid, and running tests in parallel on multiple browsers. • Completed mandatory Tekstac hands-on: Git Config, Clone Repo, Add/Commit/Push, Pull and Merge, Merge Resolve Conflict, Git Tags, Build Maven Project, Pipeline compile and test, Build Web Application. • Completed additional hands-on: GIT installation and commands implementation, Install Jenkins and Creation of new job.			

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Fortnightly Progress Report

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1. Company Overview:

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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 11, I interacted with Cognizant Academy trainers and fellow GenC trainees during Master Sessions on Automation Framework design and DevOps tools. All sessions were conducted within the Training division. The week covered Day 46 (Automation Framework) and Days 47 to 51 (DevOps – GIT, Maven, Grid, Jenkins), with hands-on exercises being the primary activity alongside daily trainer-led sessions.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Automation Framework Design (Day 46): Attended the Master Session on designing a complete Hybrid Automation Framework combining POM, Data-Driven, and Keyword-Driven approaches. Studied Test Data Handling using Excel/CSV/XML/JSON, Extent Reports, Log4j logging, and screenshot capture strategies for failure analysis.

Activity 2 – GIT & Maven Hands-On (Days 47–51): Practiced core GIT commands (clone, add, commit, push, pull, merge, resolve conflicts, tags) on the Tekstac platform. Also learned Maven project structure, dependency management via pom.xml, Maven build lifecycle, and executing Selenium tests using Maven goals from the command line.

Activity 3 – Jenkins CI & Selenium Grid Setup: Configured Jenkins CI server, created a Jenkins Pipeline integrating GIT and Maven, and set up job scheduling for automated nightly test runs. Also studied Selenium Grid Hub and Node architecture and practiced parallel test execution across multiple browsers.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Faced difficulty resolving merge conflicts in GIT when multiple branches had changes to the same file. Practiced the Merge – Resolve Conflict hands-on on the Tekstac platform and referred to BU Chapter 17 content for guidance. This experience taught the importance of frequent commits and clear branching strategies to minimize merge conflicts in team-based development environments.



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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with GIT version control system and GitHub for remote repository management, Maven for Java project build automation and dependency management, Jenkins for CI/CD pipeline creation and test scheduling, and Selenium Grid for distributed parallel test execution. Also studied BU Training Content Chapters 17 to 19 covering Selenium Grid, Maven, GIT Version Control, and Jenkins CI Systems with their PPT and video reference content.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

A dedicated hands-on session showing the complete end-to-end CI/CD pipeline – from a developer pushing code to GIT, Maven building the project, and Jenkins automatically triggering the Selenium test suite and generating reports – would greatly help trainees understand how all these DevOps tools work together in a real enterprise pipeline rather than learning each tool in isolation.

7. What did you find novel/exciting/valuable during this period?

Setting up a Jenkins pipeline that automatically triggers Selenium tests whenever code is pushed to GitHub was the most exciting experience of this week. Seeing the entire CI/CD workflow – from code commit to test execution to report generation – happen automatically without any manual intervention gave a real sense of how modern DevOps-driven quality engineering works in enterprise projects at Cognizant.

8. What according to you, can be of use to your institute and students and in what ways?

Introducing GIT, Maven, and Jenkins as part of the academic software engineering or final-year project curriculum would be immensely valuable. Most students submit projects without any version control, build automation, or CI practices. Even a basic lab session covering GIT branching, Maven project setup, and Jenkins job creation would make students significantly more industry-ready compared to their peers.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 11 were completed. The Automation Framework session (Day 46), mandatory and additional Tekstac hands-on exercises for GIT and Maven (Days 47–51), BU Chapter content (Chapters 17–19), Udemy Section 8 (CI – Maven, Git, GitHub & Jenkins), and the Integrated Project deliverable for Scalable Execution Infrastructure were all progressed within the scheduled time.

Payal SamarKumar Sahu

Name and Signature of Student.

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering (IoT)****STUDENT'S DIARY/ LOG****Your Name:** Payal SamarKumar Sahu**Your Institute:** Yeshwantrao Chavan College of Engineering Nagpur**Company where internships are carried out :** Cognizant Technology Solutions Pvt. Ltd**Period:** From: 31/01/25 To: 13/05/26

Week: 12		DATE: From	20/04/26 to 24/04/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Stage 2 – DevOps (Maven/Grid), BDD Cucumber & API Automation with Postman & REST Assured (Days 52–58)	
• Completed DevOps hands-on: JUnit with Maven, Maven Directory Structure, Maven Shade Plugin, Maven App to access External Service, Capital Service, Setup Grid and parallel execution (Days 52–54). • Studied BDD Cucumber framework: Introduction to BDD and Gherkin syntax, writing Feature files and Step Definitions, integrating BDD with Selenium Framework, running scenarios and hooks (Days 55–56). • Completed mandatory Tekstac hands-on for Cucumber: Address Book and Customer Registration scenarios. • Learned POSTMAN tool: creating workspaces, collections, sending GET/POST/PUT/PATCH/DELETE API requests, environment variables, data parameterization using CSV, and testing APIs using Swagger UI. • Studied REST Assured API Automation: basic API fundamentals, verifying status codes (200, 301, 404, 500), extracting JSON/XML responses, query parameters, path parameters, and base URI configuration. • Completed mandatory Tekstac hands-on for POSTMAN: User Service 1/2/3, Cinema Hall Service, Find Winner SOAP Webservice. • Completed REST Assured hands-on: verifying Bitcoin rate status codes, printing Bitcoin rates against USD/GBP/EUR currencies. • Submitted Integrated Project Deliverable: BDD Framework Integration and CI/CD Pipeline Integration (due Day 54). • Continued Hackathon project integration and worked on daily basis as per project schedule. • Continued GenAI Code Assist learning: GitHub Copilot for Beginners and GitHub Copilot Complete Guide 2025.			

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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 12, I interacted with Cognizant Academy trainers and fellow GenC trainees during Master Sessions on DevOps tools, BDD Cucumber, and API Automation. All sessions were within the Training division. Days 52–54 focused on completing DevOps Maven and Grid hands-on, Days 55–56 covered BDD Cucumber, and Days 57–58 began the API Automation module with Postman and REST Assured fundamentals.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – BDD Cucumber Framework (Days 55–56): Attended Master Sessions on Behaviour Driven Development using Cucumber. Learned Gherkin syntax for writing Feature files (Given-When-Then), creating Step Definition classes, and integrating Cucumber with the existing Selenium framework. Completed Cucumber hands-on: Address Book and Customer Registration scenarios.

Activity 2 – POSTMAN API Testing (Day 57): Learned to use POSTMAN for manual API testing. Created workspaces and collections, sent all HTTP request types (GET, POST, PUT, PATCH, DELETE), configured environment variables, practiced data parameterization with CSV files, and tested APIs using Swagger UI. Completed all five mandatory POSTMAN hands-on exercises.

Activity 3 – REST Assured API Automation (Day 58): Began REST Assured automation by studying API fundamentals and verifying HTTP status codes (200, 301, 404, 500) programmatically. Practiced extracting JSON and XML responses, using query and path parameters, and setting base URI configurations in the REST Assured framework.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Initially found Gherkin syntax and the mapping between Feature file steps and Step Definition methods confusing, especially when step parameters were involved. Referred to Udemy Section 9 (Cucumber BDD Framework) and practiced the Address Book scenario step by step. This clarified how @Given, @When, and @Then annotations map to feature file statements, reinforcing the importance of precise syntax in BDD frameworks.



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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with POSTMAN as a tool for manual and automated API testing, including workspace management, collection runners, environment variable configuration, and CSV-based data parameterization. Also explored REST Assured as a Java library for API test automation, studying its fluent API syntax for building request specifications, validating response codes, extracting JSON/XML data, and handling authentication. Referred to the Udemy course 'Webservices API Testing with Rest Assured API & POSTMAN 2024' (Sections 2 and 3).

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

Having a shared internal REST API server specifically set up for training purposes would make the REST Assured hands-on exercises more consistent. Currently, some public APIs used in exercises are occasionally unavailable or return unexpected responses, which disrupts the hands-on flow. A dedicated training API with predictable responses for all HTTP methods would improve the learning experience significantly.

7. What did you find novel/exciting/valuable during this period?

Learning BDD Cucumber and writing test scenarios in plain English using Gherkin syntax was the most exciting experience of this week. The idea that non-technical stakeholders can read and understand test scenarios written in Given-When-Then format bridges the gap between business requirements and automated test scripts. This concept of living documentation makes the Cucumber framework particularly powerful for Agile projects at Cognizant.

8. What according to you, can be of use to your institute and students and in what ways?

Introducing API testing using POSTMAN as part of the academic web development or software testing curriculum would be extremely valuable. POSTMAN is freely available and widely used across the industry for both manual and automated API testing. A basic lab session on sending HTTP requests, validating responses, and using environment variables in POSTMAN would give students practical exposure to one of the most important testing tools used in modern software development.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 12 were completed. The DevOps Maven and Grid hands-on exercises (Days 52–54), BDD Cucumber Tekstac hands-on and Udemy Section 9 (Days 55–56), POSTMAN hands-on exercises (Day 57), REST Assured API fundamentals (Day 58), Integrated Project BDD and CI/CD deliverables, and daily Hackathon project integration were all completed within the scheduled time.

Psahu

Payal SamarKumar Sahu

Name and Signature of Student.

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering (IoT)****STUDENT'S DIARY/ LOG****Your Name:** Payal SamarKumar Sahu**Your Institute:** Yeshwantrao Chavan College of Engineering Nagpur**Company where internships are carried out :** Cognizant Technology Solutions Pvt. Ltd**Period:** From: 31/01/25 To: 13/05/26

Week: 13		DATE: From	27/04/26 to 01/05/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Stage 2 – REST Assured Advanced, Docker & Containers, GenAI Hands-On & Final Evaluation (Days 59–65)	
• Completed advanced REST Assured topics: response validation (Equality, List Size, Existence, Multiple Assertions), header handling, cookie validation, redirect handling, and JSON schema validation (Days 59–61). • Practiced REST Assured authentication methods: Basic Authentication, Digest Authentication, API Key Authentication, Bearer Token, and OAuth 2.0 in both Postman and REST Assured framework. • Completed REST Assured serialization/deserialization using POJO classes, Request and Response Specification builders, and payload handling (JSON file, XML file, form-data, multipart file upload). • Completed REST Assured advanced hands-on: printing university details (USA, India, Canada), validating population data, verifying authentication APIs, and dynamic data generation using Faker. • Created a Cucumber BDD Framework supporting both API and Web automation (Chapter 17 API Fundamentals). • Integrated REST Assured with TestNG/Maven framework, Jenkins Freestyle project, parallel execution, and generated Extent Reports (Chapter 18 API Fundamentals). • Studied Docker and Containers: Introduction to Docker, images and containers, running Selenium tests using Docker containers, setting up Dockerized Selenium Grid (Day 62). • Completed GenAI Comprehensive Hands-On: Chat Endpoint Basics, Tokenization & Context Limits, Prompt Optimization, Generate Edge Cases, AI-Driven BDD Generation, Dynamic Locator Generation, and AI-Augmented Framework			

Signature and stamp of Industry Supervisor



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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 13, I interacted with Cognizant Academy trainers during Master Sessions on advanced REST Assured and Docker topics (Days 59–62). On Days 63 to 65, I was interviewed by a Technical Subject Matter Expert (SME) from the Business Unit during the Final Technical Evaluation and Final Project Evaluation, conducted via video interview on the GenC Learn Platform. This was the culminating formal evaluation of the entire GenC training program.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Advanced REST Assured & API Framework Integration (Days 59–61): Completed advanced REST Assured topics including response validations, authentication methods (Basic, Digest, API Key, Bearer Token, OAuth 2.0), POJO serialization/deserialization, payload handling, and Cucumber BDD + TestNG/Jenkins integration. Completed all Chapter 7 to 18 API Fundamentals hands-on exercises.

Activity 2 – Docker & GenAI Hands-On (Day 62): Studied Docker fundamentals including images, containers, and setting up a Dockerized Selenium Grid. Completed the comprehensive GenAI Hands-On for Automation Developers covering AI-driven BDD generation, dynamic locator generation, prompt optimization, and building an AI-Augmented automation framework using GitHub Copilot.

Activity 3 – Final Project + Technical Evaluation (Days 63–65): Delivered the completed Hackathon group project for evaluation by the BU SME, demonstrating full automation framework features including POM, TestNG, Cucumber BDD, Jenkins CI/CD, and REST Assured API automation. Also appeared for the Final Technical Evaluation video interview covering all skills across the entire GenC training curriculum.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Found OAuth 2.0 token-based authentication in REST Assured challenging as it required understanding token generation, expiry, and refresh mechanisms. Referred to Chapter 9 API Fundamentals hands-on exercises and the Udemy REST Assured course for detailed guidance. This deepened understanding of modern API security patterns and how enterprise APIs secure their endpoints, which is directly applicable to real-world QA automation work at Cognizant.



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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with Docker containerization for running Selenium tests in isolated, consistent environments using Dockerized Selenium Grid. Also explored the complete REST Assured framework covering Request Specification, Response Specification, POJO-based serialization/deserialization, JSON schema validation, and parallel test execution using TestNG. Additionally, completed the GenAI Code Assist module with GitHub Copilot, learning prompt engineering techniques for generating automation test code, feature files, and edge cases using AI assistance.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

A structured capstone project spanning the entire training curriculum – integrating Java, Selenium, TestNG, Cucumber BDD, REST Assured, Jenkins CI/CD, and Docker in a single end-to-end automated testing framework for a business application – would be an ideal final deliverable. The current Hackathon project comes close but having clearer evaluation criteria and more time for refinement would make it an even stronger portfolio piece for trainees entering the BU.

7. What did you find novel/exciting/valuable during this period?

The Final Project and Technical Evaluation was the most significant milestone of the entire GenC training journey. Demonstrating a complete automation framework built over 13 weeks – covering UI automation with Selenium, API automation with REST Assured, BDD with Cucumber, CI/CD with Jenkins, and containerization with Docker – to a BU SME was an extremely rewarding experience.

8. What according to you, can be of use to your institute and students and in what ways?

The complete GenC QEA training curriculum – from Core Java and SQL through Selenium automation, API testing, CI/CD, and Docker – represents an industry-aligned learning pathway that academic institutions could model their placement preparation programs on. Specifically, introducing REST Assured for API testing,

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 13 were successfully completed. The advanced REST Assured hands-on exercises (Chapters 7–18 API Fundamentals), Docker and Selenium Grid containerization session, comprehensive GenAI hands-on exercises, Hackathon project delivery, Integrated Project final deliverables including API Smoke Suite and REST Assured Module, and the Final Project and Technical Evaluation were all completed within the scheduled program timeline.

Psahu.

Payal SamarKumar Sahu

Name and Signature of Student.

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.edu**E-mail : principal@ycce.edu, info@ycce.edu****Department of Computer Science and Engineering (IoT)****STUDENT'S DIARY/ LOG****Your Name:** Payal SamarKumar Sahu**Your Institute:** Yeshwantrao Chavan College of Engineering Nagpur**Company where internships are carried out :** Cognizant Technology Solutions Pvt. Ltd**Period:** From: 31/01/25 To: 13/05/26

Week: 14		DATE: From	04/05/26 to 08/05/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Stage 2 – REST Assured Advanced, Docker & Containers, GenAI Hands-On & Hackathon Delivery (Days 58–62)	
• Practiced REST Assured authentication: Basic Authentication, Digest Authentication, API Key, Bearer Token, and OAuth 2.0 in both Postman and REST Assured framework. • Completed POJO-based serialization and deserialization in REST Assured framework for structured request/response handling. • Studied payload handling in REST Assured: sending JSON file, XML file, form-data parameters, and multipart file upload/download. • Completed REST Assured hands-on: university details for USA/India/Canada, population validation, authentication APIs, and dynamic data generation using Faker (Day 61). • Created Cucumber BDD Framework supporting both API and Web automation; integrated REST Assured with TestNG/Maven and Jenkins Freestyle project. • Studied Docker and Containers: Introduction to Docker, Docker images and containers, running Selenium tests using Docker containers, setting up Dockerized Selenium Grid for consistent environments (Day 62). • Completed GenAI Comprehensive Hands-On: Chat Endpoint Basics, Tokenization and Context Limits, Prompt Optimization, Generate Edge Cases, AI-Driven BDD Generation, Dynamic Locator Generation, and AI-Augmented Framework. • Submitted all remaining Integrated Project Deliverables: API Smoke Suite Postman Collection, REST Assured Automation Module, Comprehensive Automation Strategy and Consolidated Execution Reports (due Day 61).			

Signature and stamp of Industry Supervisor



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Fortnightly Progress Report

(To be mailed every alternate Monday to Project Guide, designated person(s) of the company, your HoD and the concerned Dean from your institute)

1. Company Overview:

Cognizant Technology Solutions is a leading global IT services and consulting company headquartered in the USA, with a major delivery presence across India. It specializes in digital transformation, cloud computing, AI, automation, software engineering, and quality assurance. Cognizant's QEA (Quality Engineering and Assurance) division focuses on intelligent and automated quality assurance, modernisation assurance, and experience assurance for enterprise clients across healthcare, finance, and technology sectors.

2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 14, I interacted with Cognizant Academy trainers during Master Sessions on advanced REST Assured topics and Docker containerization (Days 58–62). On Day 62, I delivered the completed Hackathon project to the BU SME for review and incorporated feedback. All interactions were within the Training division. This was the final week of technical learning content before the Final Evaluation week.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Advanced REST Assured (Days 58–61): Completed all remaining REST Assured chapters covering authentication methods (Basic, Digest, API Key, Bearer Token, OAuth 2.0), POJO serialization/deserialization, Request/Response Specification builders, payload handling (JSON, XML, form-data, multipart), and parallel test execution with TestNG. Also created a Cucumber BDD framework supporting both API and web automation and integrated it with Jenkins.

Activity 2 – Docker & Containers (Day 62): Attended the Master Session on Docker fundamentals including Docker images and containers, running Selenium tests inside Docker containers, and setting up a Dockerized Selenium Grid for consistent cross-environment test execution. Also completed the comprehensive GenAI Hands-On exercises covering AI-driven BDD generation, dynamic locator generation, and AI-Augmented framework building using GitHub Copilot.

Activity 3 – Hackathon Project Final Delivery: Delivered the completed group Hackathon project to the BU SME on Day 62 for evaluation. The project demonstrated the complete automation framework including Selenium WebDriver, POM, TestNG, Cucumber BDD, REST Assured, Jenkins CI/CD, and Docker integration. All integrated project deliverables including the API Smoke Suite, REST Assured Automation Module, and Consolidated Execution Reports were also submitted.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Faced challenges in setting up the Dockerized Selenium Grid with the correct node configurations for different browser versions. Referred to BU Chapter 17 Docker content and the Udemy Section 13 Docker material for detailed configuration steps. This experience highlighted the importance of containerization in ensuring consistent



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test execution environments across development, testing, and CI/CD pipeline stages.

5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with Docker containerization for creating isolated, reproducible test execution environments using Dockerized Selenium Grid. Also completed the comprehensive GenAI Hands-On module covering prompt engineering techniques with GitHub Copilot for generating automation test code, BDD feature files, and test edge cases. Studied the complete REST Assured framework covering serialization, authentication, and parallel API test execution integrated with Maven and Jenkins CI/CD.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

A guided walkthrough session showing how to containerize a complete Selenium automation framework – from writing the Dockerfile to running the test suite on a Dockerized Grid and publishing reports – would be very beneficial. Currently the Docker module is covered at a conceptual level and more hands-on time with Docker Compose for multi-container Selenium Grid setups would greatly strengthen practical understanding before the Final Evaluation.

7. What did you find novel/exciting/valuable during this period?

The GenAI Comprehensive Hands-On using GitHub Copilot was the most exciting experience of this week. Watching GitHub Copilot generate complete Selenium test scripts, BDD feature files, and dynamic XPath locators from plain English prompts demonstrated how AI is transforming the future of test automation.

8. What according to you, can be of use to your institute and students and in what ways?

Introducing Docker as part of the academic DevOps or cloud computing curriculum would prepare students for modern software delivery practices. Understanding how to containerize applications and test environments is a highly sought-after skill in the industry. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 14 were completed. The advanced REST Assured hands-on exercises (Chapters 7–18 API Fundamentals), Docker and Selenium Grid containerization session, comprehensive GenAI hands-on exercises, Hackathon project final delivery to BU SME, and all remaining Integrated Project deliverables including API Smoke Suite, REST Assured Automation Module, and Consolidated Execution Reports were all completed within the scheduled program timeline.

Psahu.

Payal SamarKumar Sahu

Name and Signature of Student.

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.edu[E-mail : principal@ycce.edu](mailto:principal@ycce.edu), info@ycce.edu**Department of Computer Science and Engineering (IoT)****STUDENT'S DIARY/ LOG****Your Name:** Payal SamarKumar Sahu**Your Institute:** Yeshwantrao Chavan College of Engineering Nagpur**Company where internships are carried out :** Cognizant Technology Solutions Pvt. Ltd**Period:** From: 31/01/25 To: 13/05/26

Week: 15		DATE: From	11/05/26 to 13/05/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Final Project Evaluation + Final Technical Evaluation & Program Completion (Days 63–65)	
• Prepared the final Hackathon group project demo covering complete automation framework with all integrated features. • Appeared for Final Technical Evaluation (FTE) via video interview on the GenC Learn Platform – conducted by a BU Tech SME covering all skills across the entire GenC training curriculum (Day 63). • Appeared for Final Project Evaluation (FPE) via video interview – BU SME evaluated the Hackathon group project based on Source Code, Functionality Completion, Usage of Features, Code Quality, and Demo of Output (Day 64). • Final Evaluation outcome recorded as RAG (Red/Amber/Green) Performance Health Status on the GenC Learn Platform. • Trainees with RED or AMBER status on Attempt 1 were given one additional attempt (Attempt 2) for the Final Evaluation. • Successfully completed the entire GenC QEA (Quality Engineer with Selenium Automation – v3) training program. • Received Performance Health Status (PHS) assessment covering all learning components: Hands-On, Code Challenges, ICTs, Interim and Final Evaluations. • Completed the full 65-day GenC learning journey from Core Java and SQL through Selenium Automation, API Testing, CI/CD, and Docker containerization.			

Signature and stamp of Industry Supervisor



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Fortnightly Progress Report

(To be mailed every alternate Monday to Project Guide, designated person(s) of the company, your HoD and the concerned Dean from your institute)

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2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 15, I interacted with Technical Subject Matter Experts (SMEs) from Cognizant's Business Unit during the Final Technical Evaluation and Final Project Evaluation conducted via video interview on the GenC Learn Platform on Days 63 and 64. This was the most significant professional interaction of the entire training program, representing a formal performance assessment by industry experts from the BU who will determine readiness for deployment into live projects.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Final Technical Evaluation (FTE) – Day 63: Appeared for the Final Technical Evaluation video interview conducted by a BU Tech SME on the GenC Learn Platform. The evaluation covered all skills learned throughout the 65-day GenC program including Core Java, SQL, Functional Testing, Selenium WebDriver, POM, Apache POI, TestNG, BDD Cucumber, REST Assured, DevOps tools, and Docker. Explained concepts clearly and demonstrated technical proficiency across all modules.

Activity 2 – Final Project Evaluation (FPE) – Day 64: The Hackathon group project was presented and evaluated by the BU SME. The evaluation criteria included Source Code quality, Functionality Completion, Usage of Framework Features, Code Quality, and Live Demo of Output. The project demonstrated a complete enterprise-grade automation framework integrating Selenium, TestNG, Cucumber BDD, REST Assured, Jenkins CI/CD, and Docker.

Activity 3 – Program Completion & PHS Assessment: Received the final Performance Health Status (PHS) on the GenC Learn Platform reflecting the cumulative performance across all learning components including Hands-On exercises, Code Challenges, ICT Assessments, Interim Evaluation, and Final Evaluation. Successfully completed the full 65-day GenC QEA (Quality Engineer with Selenium Automation – v3) training program.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

During the Final Technical Evaluation, faced a challenging question on optimizing TestNG parallel execution configuration in testng.xml for large test suites. Drew upon hands-on experience from the TestNG module and



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Integrated Project work to provide a structured answer. This reinforced the importance of not just completing exercises but deeply understanding the reasoning behind each configuration choice, as BU SMEs test conceptual depth and not just surface-level knowledge.

5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Revised and consolidated knowledge of the complete GenC QEA automation framework including the Hybrid Framework (POM + Data-Driven + Keyword-Driven), TestNG suite management, Cucumber BDD integration, REST Assured API automation, Jenkins CI/CD pipeline, Selenium Grid parallel execution, and Docker containerization.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

The Final Evaluation preparation could be improved by providing trainees with a structured revision checklist covering all topics from Stage 1 and Stage 2 with key concepts highlighted for each module. Currently, trainees must self-organize their revision across 65 days of content.

7. What did you find novel/exciting/valuable during this period?

Successfully completing the Final Project and Technical Evaluation and receiving the Performance Health Status on the GenC Learn Platform was the most meaningful milestone of the entire training journey. Demonstrating a fully functional enterprise-grade automation framework to a BU SME – built entirely from scratch over 13 weeks of learning – and being evaluated as deployment-ready is an achievement that reflects the comprehensive and rigorous nature of the Cognizant GenC QEA training program.

8. What according to you, can be of use to your institute and students and in what ways?

The entire GenC QEA training structure – a 65-day intensive program combining self-paced learning, ILT Master Sessions, daily hands-on exercises, Code Challenges, ICT Assessments, and formal BU evaluations – is an excellent model for industry-academia partnerships.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 15 were successfully completed. The comprehensive revision of all Stage 1 and Stage 2 topics, Hackathon project final preparation, Final Technical Evaluation, Final Project Evaluation, and the complete 65-day GenC QEA (Quality Engineer with Selenium Automation – v3) training program were all completed within the scheduled program timeline ending on 13/05/26.

Psahu

Payal SamarKumar Sahu

Name and Signature of Student.